Which sublayer runs the deduction step?

Zero ONE write at a time -- block L's MLP, or block L's attention -- on the same 350 counterfactuals as the head page.

NOT ANSWERED. No single write carries it, and the ones that do matter also break the model's ordinary prediction.

20 OF 24 WRITES ARE DISPENSABLE in the SOLVER. Every MLP from L0 to L10 keeps 0.97-1.01 of the hint-edit effect; deleting any one of them changes nothing.

THE 4 THAT ARE NOT are the read-in and the write-out, not the middle: L0 attention 0.021, L11 MLP 0.161, L10 attention 0.670, L9 attention 0.896. Each loses the model's CLEAN decision margin by the same fraction (15.5 logits -> 0.3 / 2.4), which the pre-registered sanity control says is a broken forward pass, not a deleted step.

THE ONE POSITIVE RESULT IS REDUNDANCY. At L0 the four attention heads individually keep 1.00, 0.84, 1.00, 1.00 -- no single head matters -- yet deleting all four at once keeps 0.021. Head ablation UNDERSTATES dependence, and a method that removes one component at a time cannot see a step computed in two places at once.

Job 50151317, 2 min 41 s on one GPU. SOLVER = leak-free, solves 100.0% of held-out graphs, 350 cases, all of which flip when the forcing is redirected. LEAK-READER = the leaked-encoding twin, 250 cases. Deck built from the run's CSVs; nothing typed in.

The intervention, and what was predicted before it ran

THESE ARE PRE-NORM BLOCKS, so each block makes two separate additive writes into the residual stream:

$$x \leftarrow x + \text{attn}(\text{norm}_1(x)) \qquad x \leftarrow x + \text{MLP}(\text{norm}_2(x))$$

Zeroing linear2's weight and bias makes block L's MLP write exactly zero; zeroing out_proj's does the same for its attention. The skip connection is untouched, so every later block still gets a well-formed residual stream -- this deletes one write, it does not cut the block out. Score: the fraction of the one-token hint-edit effect that survives, 1.00 = that write is irrelevant, 0.00 = the effect is gone. Same normalisation as the per-head page, so heads and whole writes can be read on one axis.

PRE-REGISTERED IN THE SBATCH HEADER, BEFORE THE RUN:

(a) at least one MLP falls below 0.60 -- below the worst single head's 0.839

CONFIRMED, and uninformative

(b) the load-bearing MLPs sit in L6-L10, where the SAE finds the forcing set

FALSIFIED -- the minimum is at L11, the named falsifier

(c) the LEAK-READER's profile differs from the SOLVER's

SPLIT -- the last MLP is shared, the read-in is not

(d) SANITY: if the clean prediction dies too, it is damage, not localisation

HELD at 20 of 24, and it is why this page is a null

Pre-registrations quoted from scripts/slurm/sudoku2_sublayer.sbatch lines 41-70. (a) is confirmed by L11's MLP, which is the output write and is disqualified by (d); (b)'s falsifier was written as 'the minimum sits at L0-L2 or at L11'.

What the model actually sees

Every token is one categorical id into a 34-symbol vocabulary (N=24, q=4, degree=5). No text, no digits-as-characters.

VOCABULARY

0-5	BEGIN_HINTS BEGIN_GRAPH BEGIN_REASONING
	BEGIN_ROUND SOLVED UNRESOLVED
	markers (UNRESOLVED = DD got stuck; unused here)
6-9	COLOR_0 COLOR_1 COLOR_2 COLOR_3
	the q colours (token name COLOR_n, prose 'colour')
10-33	N0 N1 ... N23
	one id per node, no structure in the id

GRAMMAR

BEGIN_HINTS	(NODE COLOR)*
BEGIN_GRAPH	(NODE (NODE){degree})*
BEGIN_REASONING	(BEGIN_ROUND (NODE COLOR)+)*
SOLVED	

*regex: * = zero or more, + = one or more, {degree} = exactly 5*

The graph section is a neighbour table, not an image: node, then its 5 neighbours.

THE ORDER WITHIN A ROW IS THE LEAK.

THIS INSTANCE, 198 TOKENS

HINTS	BEGIN_HINTS N2 COLOR_2 N3 COLOR_2 N5 COLOR_2 N8 COLOR_3 N10 COLOR_0 N11 COLOR_1 ... (31 tokens)
GRAPH	BEGIN_GRAPH N0 N16 N5 N18 N13 N14 N1 N8 N13 N2 N11 N19 ... (145 tokens)
REASONING	BEGIN_REASONING BEGIN_ROUND N0 COLOR_1 N1 COLOR_0 N4 COLOR_0 N6 COLOR_1 N9 COLOR_1 N12 ... (22 tokens)

TRAINING LOSS is next-token cross-entropy on EVERY token, the problem statement included -- there is no section mask.
AT EVAL the model is PROMPTED with everything through BEGIN_REASONING and generates the rest: 2 rounds, 9 deductions,
18 node+colour tokens, plus the BEGIN_ROUND markers and the SOLVED/UNRESOLVED terminator, which are trained targets like any other token.

DD is single-candidate elimination -- sudoku's 'naked single' rule transplanted to graph colouring, which is where the code's sudoku names come from; the task itself is the graph, not a 9x9 grid. Instance shown: seed 50000000004.

One real reasoning trace

15 hints of 24 nodes, then 2 rounds of simultaneous deduction. Hints: N2=COLOR_2 N3=COLOR_2 N5=COLOR_2 N8=COLOR_3 N10=COLOR_0 N11=COLOR_1 ...

THE TRACE, AS EMITTED

```
BEGIN_ROUND            N0 COLOR_1 N1 COLOR_0 N4 COLOR_0 N6 COLOR_1 ...
  8 deduced
BEGIN_ROUND            N7 COLOR_2
  1 deduced
SOLVED
```

WHY ONE DEDUCTION FIRES

node N0, neighbours N16 N5 N18 N13 N14

their colours known before this round:

```
   [0, 2, 3]  (3 = q-1 of the 4)

   (4 known neighbours, 3 DISTINCT colours -- some repeat)

   still unknown: N16
```

only colour left: 1

```
so the trace emits  N0 COLOR_1
```

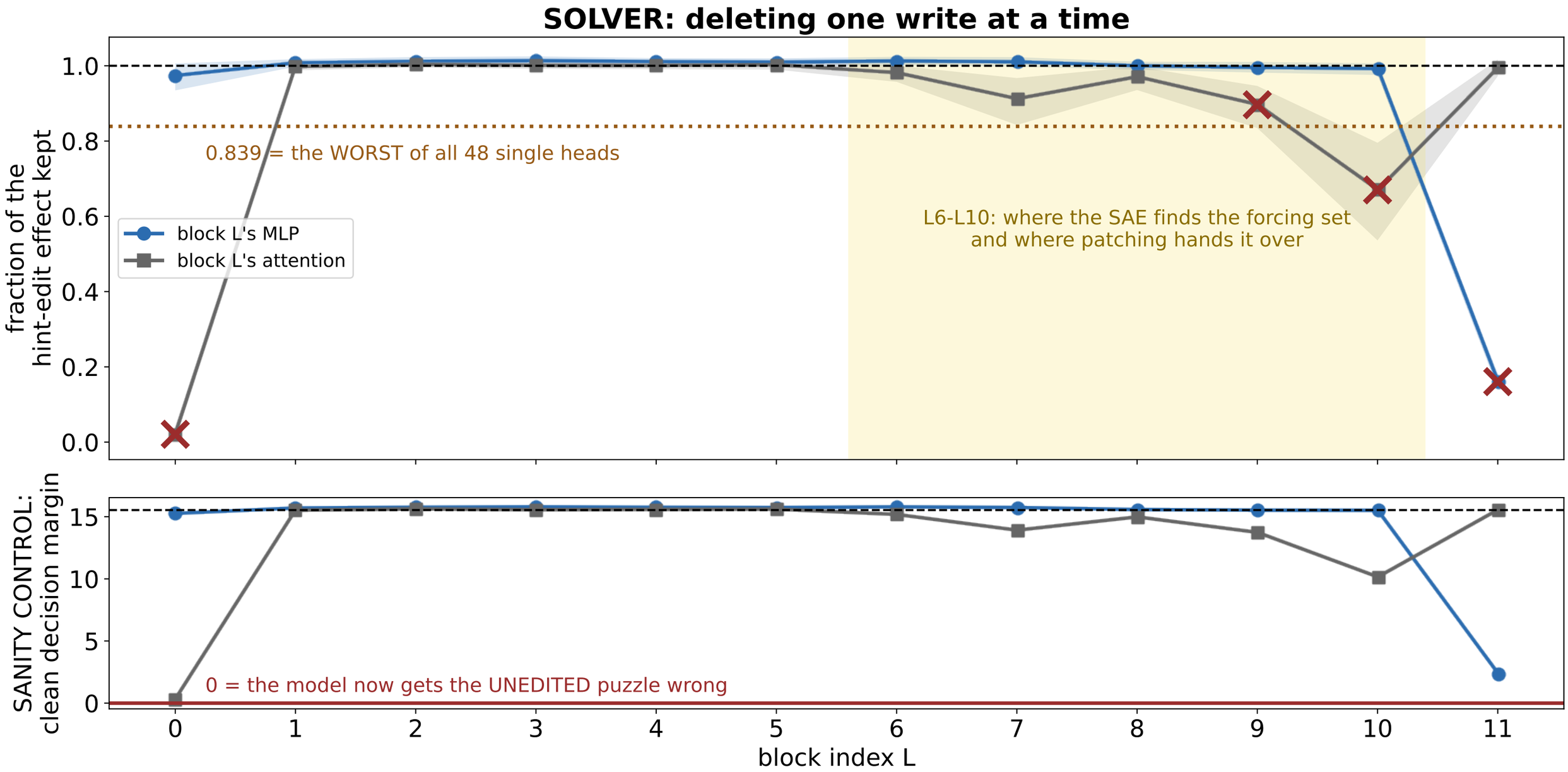
The model is never shown this reasoning --
only the node/colour pair.

A round deduces every node whose neighbours already cover q-1 colours, all at once. Reasoning depth T is the number of rounds: a deduction in round 3 depends on ones made in rounds 1 and 2, which is the serial computation that network depth has to supply.

The hint-edit counterfactual changes ONE colour token in the HINTS section; this run then deletes one sublayer write and asks how much of the model's respon

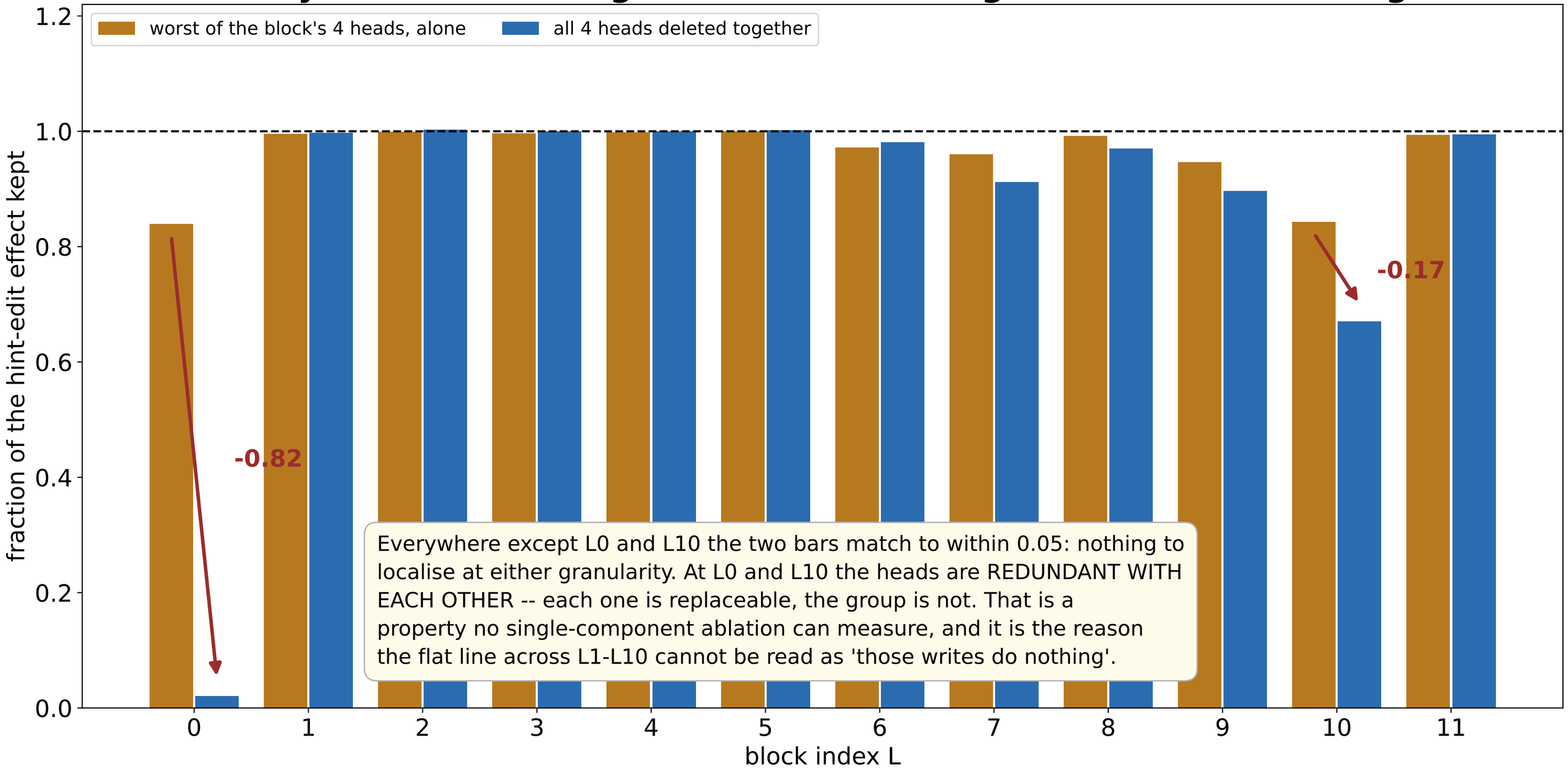
T=2 rounds, the median of 40 draws from this config (range 1-6) -- one instance, shown to make the task legible; no result in this deck is computed from it. The worked deduction on the right uses the state BEFORE this round, which is what DD itself saw; it is recomputed from the graph, not read out of the model.

The effect and the clean prediction die in the same places, by the same fractions



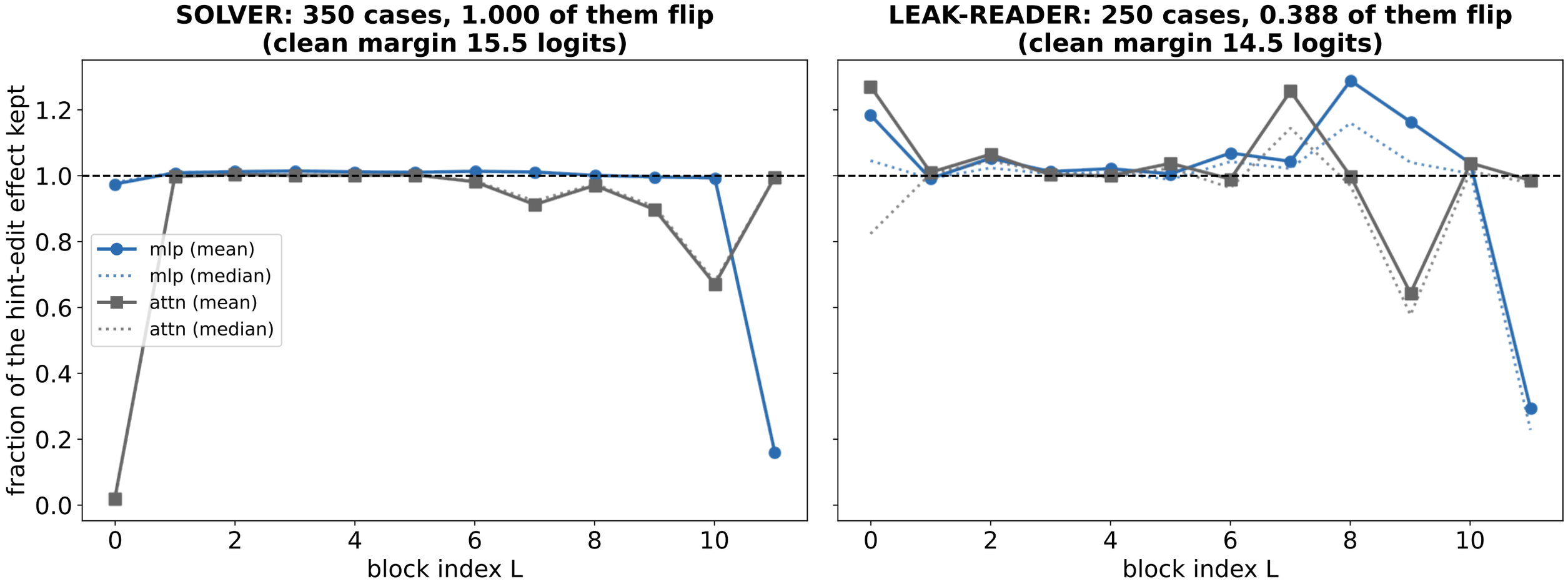
Red crosses mark the four writes below 0.90. Shaded band is the 10th-90th percentile over the 350 cases. A write that ran the deduction and nothing else would drop in the top panel and stay flat in the bottom one; none does.

Only two blocks disagree -- and that disagreement is the finding



Same 350 counterfactuals, same normalisation, both runs on the SOLVER checkpoint: head numbers from sudoku_circuits_leakfree_2026-10-28 (finding 27), whole-write numbers from this run. 3 of 48 heads fall below 0.95; 4 of 24 writes do.

The control splits: the last MLP is shared, the READ-IN is not



L0 attention: 0.021 in the SOLVER, 1.270 in the LEAK-READER (median 0.823). L11 MLP: 0.161 against 0.295 -- shared, and it is the output write in both. The LEAK-READER flips on only 0.388 of its cases, so its ratios divide two small numbers and the median is the number to trust; its apparent amplification above 1.0 shrinks from 1.29 to 1.16 and should be read as 'not needed, and noisy'.

The LEAK-READER reads the colouring off the adjacency ORDER (finding 16), so a hint edit changes the deduced answer but not what it reads -- which is why it flips on 39% of the cases where the SOLVER flips on 100%.

What this measures, and what it does not

A null result with a named reason, not a missing one

WHAT THIS RULES OUT

A single MLP that runs the elimination step. Every MLP from L0 to L10 is individually dispensable, including L8, where the dictionary finds the forcing-set features and where patching hands the fact over. The correlational and causal localisations do NOT agree at the level of one component.

WHAT IT CANNOT RULE OUT

A step computed redundantly in two or more places -- which is not hypothetical here: the L0 heads demonstrate exactly that pattern one level down. A flat 1.00 is equally consistent with 'irrelevant' and with 'duplicated'.

METHOD CAVEATS

One seed, one checkpoint per arm, one intervention type. Zeroing a weight pushes the block off-distribution rather than removing its information; mean- or resample-ablation is the gentler test. Scores are means of per-case ratios, which is solid for the SOLVER and soft for the LEAK-READER.

NEXT TEST

Joint ablation: pairs of writes, and whole L6-L10 windows at once, scored the same way on the same 350 cases. If redundancy is the explanation, a window drops where none of its members does.

The DD verdict (findings 27, 28) rests on the three behavioural interventions, which do move the answer: redirect the forcing and the SOLVER flips on 350/350; break it and it withholds the node 350/350 against a 0.94 placebo ceiling. This page is about WHERE, which is still open.